P Pranav

pranavrobotics.eng@gmail.com | +91 9629091899 | Chennai, Tamil Nadu
ppranav04.github.io | linkedin.com/in/ppranav04 | github.com/ppranav04

Technical Skills

Languages: C++, Python, MATLAB
Robotics: ROS 2 (Humble/Jazzy), RViz, Gazebo, URDF, RTAB-Map, A*/Dijkstra
Perception & Learning: OpenCV, YOLOv8, imitation learning (ACT), PyTorch, LeRobot
Tooling: Git, Docker, Linux
Hardware: Nvidia Jetson, Raspberry Pi, Arduino, ESP32, Intel RealSense

Work Experience

Indira Gandhi Centre for Atomic Research (IGCAR), Kalpakkam **Jan 2026 - Apr 2026**
Bachelor Thesis Intern

- Implemented a **visibility graph + A* path planner for a SCARA manipulator**, running natively on the PLC motion controller and validated in real time on a 1:1 hardware test rig.
- Benchmarked A* against Dijkstra across 1-10 obstacles resulting in identical optimal path lengths, with A*'s compute-time advantage growing to 48% at 10 obstacles.
- Technologies used: Codesys, A*, Dijkstra.

Zoho Corporation, Chennai **Sep 2024 - Feb 2025**
IoT Engineer Intern

- Built an **ESP32 air-quality monitor** in embedded C++ reading 3 gas sensors, converting raw ADC readings to ppm through power-law calibration curves derived from the datasheets.
- Delivered a working prototype at ₹10k per unit against ₹20k-40k for commercial monitors, with OTA updates and alerts through Zoho Cliq and Mail.
- Technologies used: C++, ESP32, OTA updates, MQ-series gas sensors

IGCAR, Kalpakkam **Jun 2024 - Jul 2024**
Intern Trainee

- Developed an **iterative Jacobian IK for a SCARA manipulator** in Python, adding speed-based time interpolation to cap joint velocities and remove the step changes in velocity that would trip the motors.
- Built an obstacle-avoidance planner with an interactive Pygame simulator, later extended into the multi-obstacle thesis planner.
- Technologies used: Python, NumPy, SciPy, Matplotlib, Pygame.

Education

Vellore Institute of Technology, Chennai **2022 - 2026**
B.Tech. in Mechatronics and Automation **CGPA: 8.61/10**
Relevant Coursework: Python, C++, Machine Vision, Control Systems, Artificial Intelligence, Industrial Robotics.

Academic Projects

BT-101: Autonomous Manipulation Platform (github.com/ppranav04/BT-101) Independent, Ongoing

- Rebuilt joystick and leader-follower teleop for a **6-DOF SO-101 robot arm** pair from scratch and profiled loops at 293/145 Hz.
- Trained an **ACT policy** on 15 teleop demos with a single-camera setup and deployed it on the SO-101 arm.
- The policy reproduced approach-grasp-place but missed grasps; isolated 3 testable causes (monocular depth, start coverage, no recovery data).
- Stack: Python, PyTorch, LeRobot, scservo_sdk, pygame, W&B, Hugging Face Hub

Agrirover: Vision-Guided Tomato Harvesting Arm VIT, Project-I

- Built the **perception-to-manipulation pipeline** in **ROS 2** (detection - IK - serial bridge - Arduino PWM) driving a 5-DOF 3D-printed servo arm.
- Localised fruit in 3D by deprojecting RealSense D435 depth through the camera intrinsics and a calibrated camera-to-base transform with 20.3 mm RMSE over a 0.15-0.50 m range.
- **Fine-tuned YOLOv8n** on a custom tomato-ripeness dataset: F1: 0.91-0.92 on 784 test samples.
- Stack: ROS 2, Python, C++, YOLOv8, NVIDIA Jetson Orin Nano, RealSense D435